

Data Centers What Do They Do Money Style?

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Abstract

The presence of data centers in a county correlates with changes in economic factors and worse air quality. By examining factors such as personal income per capita, PM2.5 concentration, number of employees, we have been able to predict with 80% accuracy whether a data center is present in a county. In this project, we show how and why this is significant, and what this means for the future implementation of data centers.



Introduction & Motivation

Problem & question:

In the first quarter of the 21th century, there was a rapid increase in the speed and scale of data center deployment. This project aims to understand whether this is correlated with positive or negative impacts on the local economies where these data centers are being implemented, and how these data centers may affect the lives of the average person in one of these counties.

How does the deployment of a new data center relate to local economic conditions, including employment, income, and business development patterns at the county level?

Impact & stakeholders:

The main stakeholders are people who live in a county with data centers, and those in an area that may be planning to implement new data centers in the near future. There has already been considerable pushback on data centers (such as at town hall meetings), so the findings of this project may influence whether people continue to push back against them and, ultimately, whether they are built. In which case, the stakeholders may also include those building new data centers.



Background & Related Work

Published academic research on the economic impact of data centers is currently limited, given the recency of this infrastructure's development and deployment. Michael Hicks self-published a well-researched paper titled Data Centers and Local Job Creation¹, which focuses exclusively on Texas Data Center developments and their influence on the job market at the county level. Our research investigates counties in every state within the contiguous United States, includes spatial relationships between neighboring counties, and incorporates economic variables beyond employment within the construction and information sectors.

Daniel Yue and Yiang Zeng offer one of the most complete analyses with their pre-print paper The Local Economic Effects of Data Center Entry². This text analyzes the county-level impact the entry of a new data center has on employment rates, median income, total wages, residential building activity, and business establishments. Our analysis includes all of those variables and adds air quality to better capture the negative environmental externalities associated with data centers.

The Cloud Next Door by Nagata et. al.³ establishes a link between data centers and air pollution, and Maniloff and Schreiber's Valuing Air Pollution's Impact on Labor Productivity in General Equilibrium⁴ allows us to assign a dollar value to marginal changes in PM2.5 levels.



Reproducibility & References

Repository: [https://github.com/wu-msds-capstones/project-writeup-ian_and_logan] · Key dependencies: [Python 3.13, pandas, scikit-learn, seaborn , sqlalchemy]

References: [1] Michael Hick, "Data Centers and Local Job Creation", 2025. [2] Daniel Yue and Yiang Zeng, "The Local Economic Effects of Data Center Entry", 2026 [3] Nagata et. al., "The Cloud Next Door", 2025 [4] Maniloff and Schreiber, "Valuing Air Pollution's Impact on Labor Productivity in General Equilibrium", 2024



Data, Engineering & Ethics

Sources & scale:

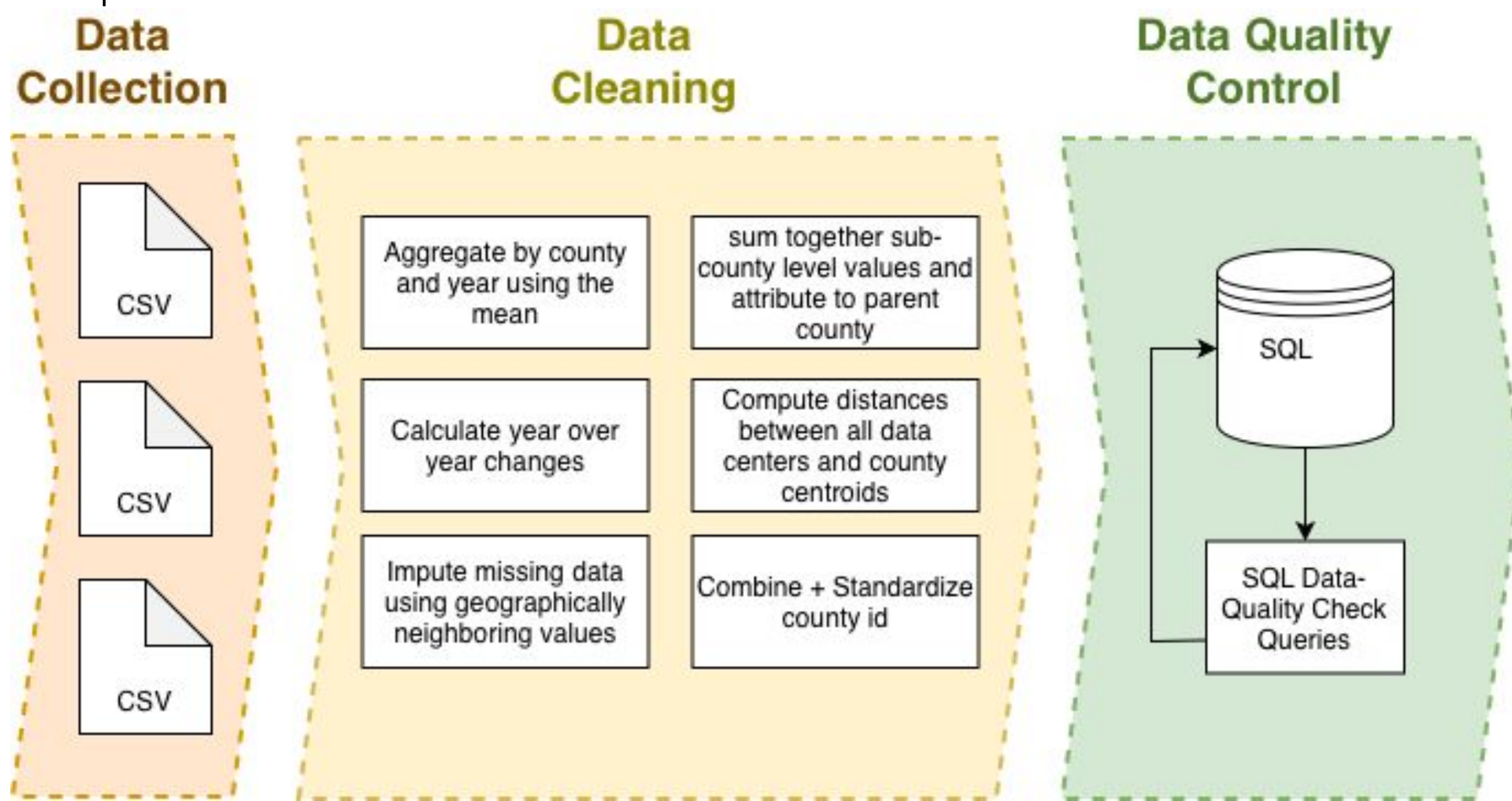
This project uses data with a granularity of county within the contiguous United States between the years 2001 and 2022. All raw data is sourced from U.S. federal government departments, including the EAP, BEA, Census, and CDC.

Pipeline:

All raw data was downloaded as a CSV. Individual cleaning scripts were written for each data source in Python. Each Python script exports a cleaned and transformed version of the data into a SQL table. Finally, we run data quality checks against the entire database using SQL queries.

Ethics & responsible use:

All of the data used for this project was pre-aggregated and provided by the government, so there is no personally identifiable or sensitive information that was used or produced. However, one area of consideration is that we are looking at economic data, which may be affected by underlying demographic variables. If a county has demographic changes that relate to both economic impacts and that also attract the construction of data centers, this may be a source of potential bias. Another inherent risk of economic data is that some of the numbers that we use may represent real major impacts on people's lives (such as unemployment), and thus should be treated with caution so as not to diminish these real-world impacts.



Methods & Evaluation

Approach:

Our initial approach was to use regressions to model data centers as both the explanatory variable (with a linear regression) and as the response variable (with a logistic regression). To help find the appropriate economic variables to include, we created some basic visualizations to understand which variables were correlated with each other and which variables were noticeably different in counties with data centers. After creating these initial regressions, we also created a random forest model to predict whether a data center was present in a county using the various economic variables that we selected. The next planned phase of our project is to update these models with better feature engineering and to try creating a neural network model. We will also need to create a model that predicts the year that a data center was created so that we can train our other models on time-based data.

Validation:

So far, we have checked statistical assumptions for all models that we've run (such as checking the LINE assumptions of a linear regression model). For our year-prediction model, we will also be checking some of the results manually to see if our model is accurately predicting the year that a data center began operation for a subset of the data. This will allow us to validate the accuracy of this model.



Results & Visualizations

Model Predicting Counties with Data Centers

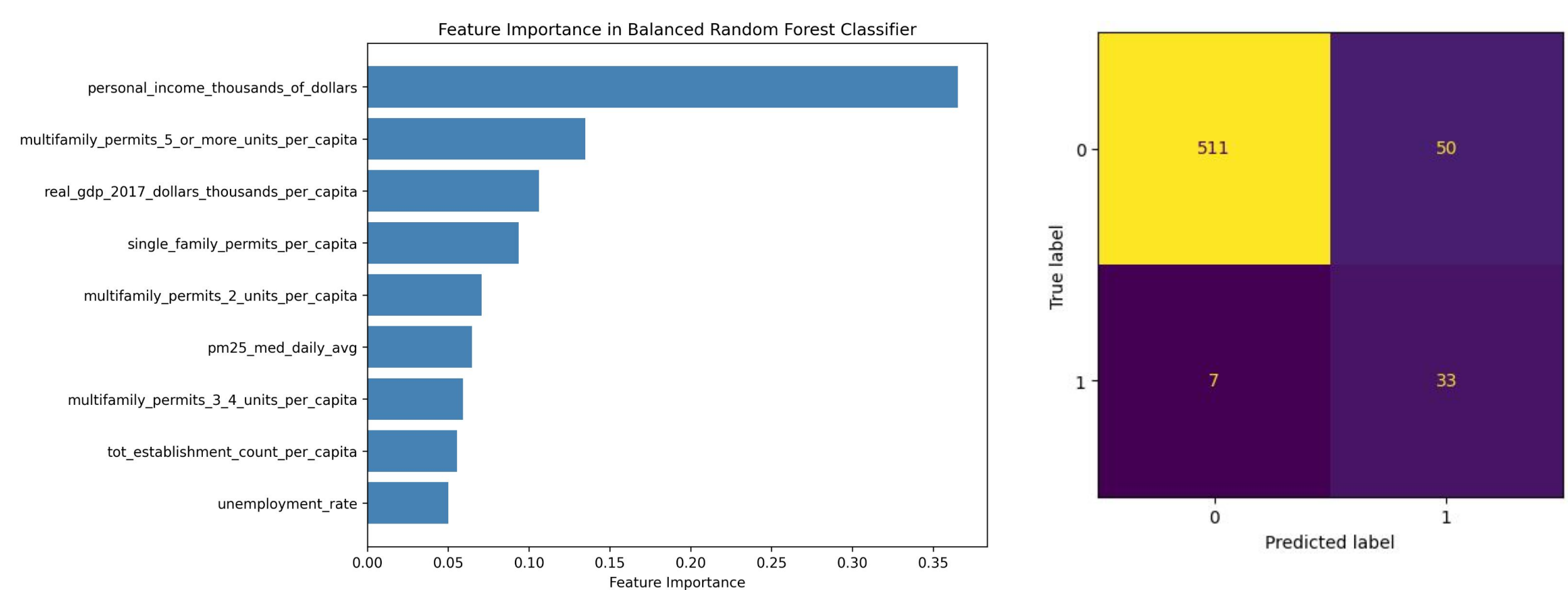


Figure 1 (DRAFT): X axis shows the relative contribution of made to decreasing the model impurity out of 100%. Y axis labels each feature in the model. Model trained on 2400 observations of counties.

Figure 2 (DRAFT): X axis shows whether the model predicted a county did (1) or did not (0) have a county. Y axis shows whether there are (1) or are not (0) datacenters in a county. Model tested on 601 observations.

Economic Impact Modeling Results

Economic Variable	Predicted Impact of Additional Data Center
Unemployment Rate	+-. ##%
Median Wage	+-. \$####
Air Pollution Induced Change in Productivity	+-. \$####
County GDP	+-. \$####

Table 1 (DRAFT): This table shows the average predicted change in a given economic variable associate with the addition of a data center in a county.



Discussion, Limitations & Conclusions

Discussion:

While the results of most of our models were not statistically significant, the random forest model that we created shows some promise. The relative success of this model does show that there is some relationship between data center presence and some of the economic factors that we looked at, but it does not mean that there is any causal relationship between data centers and economic variables.

Limitations & future work:

The biggest limitation of our work is that we were not able to obtain data on when each of the data centers became operational. Because of this, we needed to estimate these years using machine learning models, but this introduces quite a bit of uncertainty. Future research would benefit greatly from having the actual years available from the beginning.

The other major limitations surround the use of environmental factors. Although there are three major environmental areas that we hoped to look at (air, water, and land use), we were only able to obtain sufficient data for the air quality. Additionally, the translation of air data from pure PM2.5 quality to economic value requires quite a few assumptions and is not exact.

Conclusions:

The major takeaway from this project is that there *is* a relationship between data center implementation and local economic factors. Although a causal relationship cannot be established, it will be important to consider how data centers *may* relate to economic conditions when considering whether one should be built in the future.

